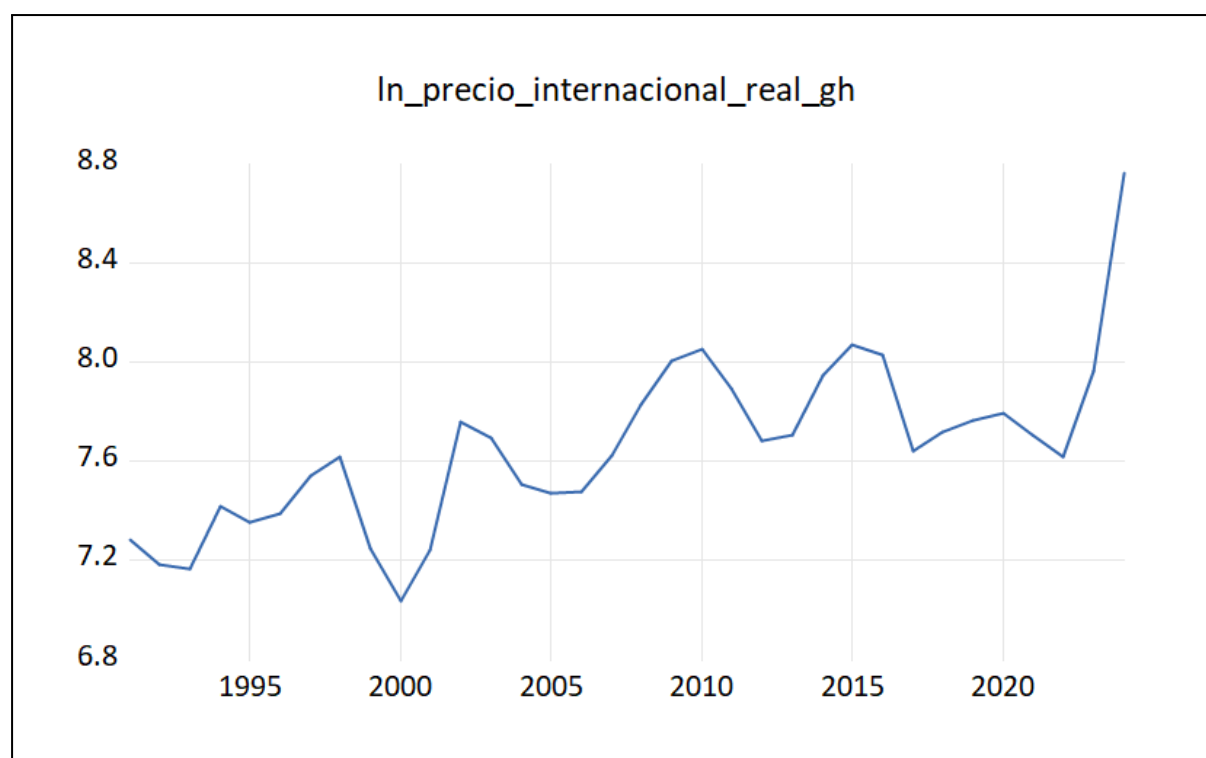




Date: 09/26/26 Time: 20:59
Sample: 1991 2024
Included observations: 34

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 0.580	0.580	12.458	0.000
		2 0.261	-0.113	15.057	0.001
		3 0.209	0.162	16.774	0.001
		4 0.303	0.200	20.524	0.000
		5 0.331	0.086	25.157	0.000
		6 0.287	0.067	28.770	0.000
		7 0.245	0.060	31.494	0.000
		8 0.281	0.125	35.215	0.000
		9 0.147	-0.203	36.277	0.000
		10 0.002	-0.113	36.277	0.000
		11 -0.053	-0.101	36.428	0.000
		12 0.020	0.001	36.450	0.000
		13 0.075	0.002	36.777	0.000
		14 0.009	-0.067	36.783	0.001
		15 -0.110	-0.073	37.563	0.001
		16 -0.174	-0.097	39.628	0.001

El correlograma no es claro, la FAC si bien se anula en el segundo rezago está muy cerca de la banda de aceptación e incluso en el 5 rezago vuelve a ser significativo, en la FACP es más claro que se anula rápidamente en el segundo rezago.

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/26/26 Time: 21:00

Sample (adjusted): 1992 2024

Included observations: 33 after adjustments

Failure to improve likelihood (non-zero gradients) after 14 iterations

Coefficient covariance computed using outer product of gradients

MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.172838	0.178031	40.28981	0.0000
@TREND	0.030318	0.009159	3.310144	0.0025
AR(1)	0.342131	0.281319	1.216168	0.2337
MA(1)	0.632968	0.203806	3.105735	0.0042
R-squared	0.705223	Mean dependent var	7.664060	
Adjusted R-squared	0.674729	S.D. dependent var	0.336927	
S.E. of regression	0.192158	Akaike info criterion	-0.347784	
Sum squared resid	1.070818	Schwarz criterion	-0.166389	
Log likelihood	9.738431	Hannan-Quinn criter.	-0.286750	
F-statistic	23.12653	Durbin-Watson stat	1.513729	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.34			
Inverted MA Roots	-.63			

Sacando el componente AR(1)

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/26/26 Time: 21:01

Sample: 1991 2024

Included observations: 34

Failure to improve likelihood (non-zero gradients) after 9 iterations

Coefficient covariance computed using outer product of gradients

MA Backcast: 1990

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.218259	0.110545	65.29727	0.0000
@TREND	0.027131	0.005684	4.773354	0.0000
MA(1)	0.722113	0.151778	4.757687	0.0000
R-squared	0.698915	Mean dependent var	7.652925	
Adjusted R-squared	0.679490	S.D. dependent var	0.338076	
S.E. of regression	0.191397	Akaike info criterion	-0.384835	
Sum squared resid	1.135619	Schwarz criterion	-0.250156	
Log likelihood	9.542200	Hannan-Quinn criter.	-0.338906	
F-statistic	35.98047	Durbin-Watson stat	1.357248	
Prob(F-statistic)	0.000000			
Inverted MA Roots	-.72			

Date: 09/26/26 Time: 21:01
Sample: 1991 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 0.155	0.155	0.8871	
		2 -0.127	-0.155	1.5068	0.220
		3 -0.329	-0.296	5.7813	0.056
		4 0.039	0.133	5.8423	0.120
		5 -0.040	-0.160	5.9103	0.206
		6 0.051	0.001	6.0219	0.304
		7 -0.181	-0.182	7.5118	0.276
		8 0.001	0.000	7.5118	0.378
		9 -0.100	-0.144	7.9999	0.433
		10 -0.104	-0.242	8.5522	0.480
		11 -0.092	-0.054	9.0021	0.532
		12 -0.032	-0.240	9.0600	0.616
		13 0.037	-0.071	9.1410	0.691
		14 0.106	-0.070	9.8338	0.707
		15 -0.019	-0.220	9.8582	0.772
		16 0.126	0.118	10.943	0.757

*Probabilities may not be valid for this equation specification.

Si bien los residuos dan ruido blanco para el modelo MA(1), en el rezago 3 está muy cerca de la banda de aceptación, por lo que incluyo los componentes MA(2) y MA(3).

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 09/28/26 Time: 15:40
Sample: 1991 2024
Included observations: 34
Failure to improve likelihood (non-zero gradients) after 29 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1988 1990

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.215131	0.095021	75.93192	0.0000
@TREND	0.026922	0.005040	5.341196	0.0000
MA(1)	1.029888	0.179285	5.744421	0.0000
MA(2)	0.065263	0.281195	0.232092	0.8181
MA(3)	-0.475698	0.164846	-2.885718	0.0073
R-squared	0.796690	Mean dependent var	7.652925	
Adjusted R-squared	0.768648	S.D. dependent var	0.338076	
S.E. of regression	0.162612	Akaike info criterion	-0.659851	
Sum squared resid	0.766834	Schwarz criterion	-0.435386	
Log likelihood	16.21746	Hannan-Quinn criter.	-0.583302	
F-statistic	28.40989	Durbin-Watson stat	1.566502	
Prob(F-statistic)	0.000000			
Inverted MA Roots	.53	-.78-.53i	-.78+.53i	

Como el componente MA(2) no es significativo lo saco de la estimación.

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 09/28/26 Time: 15:41
Sample: 1991 2024
Included observations: 34
Failure to improve likelihood (non-zero gradients) after 9 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1988 1990

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.220907	0.084041	85.92152	0.0000
@TREND	0.026518	0.004364	6.076622	0.0000
MA(1)	0.989262	0.038336	25.80473	0.0000
MA(3)	-0.513453	0.028631	-17.93323	0.0000
R-squared	0.796598	Mean dependent var	7.652925	
Adjusted R-squared	0.776258	S.D. dependent var	0.338076	
S.E. of regression	0.159915	Akaike info criterion	-0.718220	
Sum squared resid	0.767183	Schwarz criterion	-0.538648	
Log likelihood	16.20974	Hannan-Quinn criter.	-0.656981	
F-statistic	39.16370	Durbin-Watson stat	1.522242	
Prob(F-statistic)	0.000000			
Inverted MA Roots	.57	-.78-.53i	-.78+.53i	

Date: 09/28/26 Time: 15:41
Sample: 1991 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 0.146	0.146	0.7922	
		2 -0.082	-0.105	1.0474	
		3 -0.149	-0.125	1.9278	0.165
		4 -0.093	-0.062	2.2805	0.320
		5 0.016	0.016	2.2915	0.514
		6 -0.058	-0.098	2.4378	0.656
		7 -0.163	-0.169	3.6435	0.602
		8 -0.075	-0.050	3.9079	0.689
		9 -0.000	-0.033	3.9079	0.790
		10 -0.193	-0.287	5.8127	0.668
		11 -0.054	-0.064	5.9677	0.743
		12 -0.091	-0.174	6.4333	0.778
		13 0.003	-0.122	6.4338	0.843
		14 0.185	0.046	8.5262	0.743
		15 -0.033	-0.202	8.5963	0.803
		16 0.163	0.129	10.410	0.732

*Probabilities may not be valid for this equation specification.

Ambos componentes son significativos y se llega a un residuo ruido blanco, sin embargo las raíces invertidas son muy cercanas a 1 ($-.78-.53i = 0.94$), por lo que descarto este modelo como estimación de la variable.

Por lo tanto pruebo con un componente AR(1) debido que es lo que se observaba en el correlograma y puede ocurrir que al juntarlo con un componente MA no de significativo.

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/28/26 Time: 15:06				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Convergence achieved after 3 iterations				
Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.175960	0.180654	39.72215	0.0000
@TREND	0.029693	0.009205	3.225800	0.0030
AR(1)	0.527302	0.197602	2.668509	0.0122
R-squared	0.608244	Mean dependent var	7.664060	
Adjusted R-squared	0.582127	S.D. dependent var	0.336927	
S.E. of regression	0.217800	Akaike info criterion	-0.123969	
Sum squared resid	1.423108	Schwarz criterion	0.012077	
Log likelihood	5.045493	Hannan-Quinn criter.	-0.078194	
F-statistic	23.28917	Durbin-Watson stat	1.137538	
Prob(F-statistic)	0.000001			
Inverted AR Roots	.53			

Date: 09/28/26 Time: 15:07					
Sample (adjusted): 1992 2024					
Q-statistic probabilities adjusted for 1 ARMA term					
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 0.262	0.262	2.4837	
		2 -0.329	-0.427	6.5123	0.011
		3 -0.318	-0.109	10.416	0.005
		4 0.001	0.010	10.416	0.015
		5 0.098	-0.084	10.814	0.029
		6 -0.004	-0.063	10.815	0.055
		7 -0.143	-0.139	11.723	0.068
		8 0.008	0.082	11.726	0.110
		9 -0.069	-0.275	11.955	0.153
		10 -0.113	-0.080	12.595	0.182
		11 -0.088	-0.149	12.999	0.224
		12 -0.011	-0.159	13.005	0.293
		13 0.081	-0.037	13.389	0.341
		14 0.047	-0.166	13.522	0.408
		15 -0.034	-0.073	13.597	0.480
		16 0.076	-0.003	13.989	0.526
*Probabilities may not be valid for this equation specification.					

El residuo no da ruido blanco, ya que la banda del segundo rezago de la FACP da significativa, por lo que habría que incluir un coeficiente AR(2) a la estimación.

Dependent Variable: LN_PRECIO_INTERNACIONAL_REAL_GH
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)

Date: 09/27/26 Time: 11:07

Sample (adjusted): 1993 2024

Included observations: 32 after adjustments

Convergence achieved after 2 iterations

Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.247435	0.075898	95.48960	0.0000
@TREND	0.025111	0.004048	6.203496	0.0000
AR(1)	0.938365	0.186313	5.036484	0.0000
AR(2)	-0.790799	0.186325	-4.244198	0.0002
R-squared	0.746185	Mean dependent var	7.679072	
Adjusted R-squared	0.718990	S.D. dependent var	0.330916	
S.E. of regression	0.175420	Akaike info criterion	-0.526804	
Sum squared resid	0.861616	Schwarz criterion	-0.343587	
Log likelihood	12.42887	Hannan-Quinn criter.	-0.466073	
F-statistic	27.43885	Durbin-Watson stat	1.543643	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.47-.76i	.47+.76i		

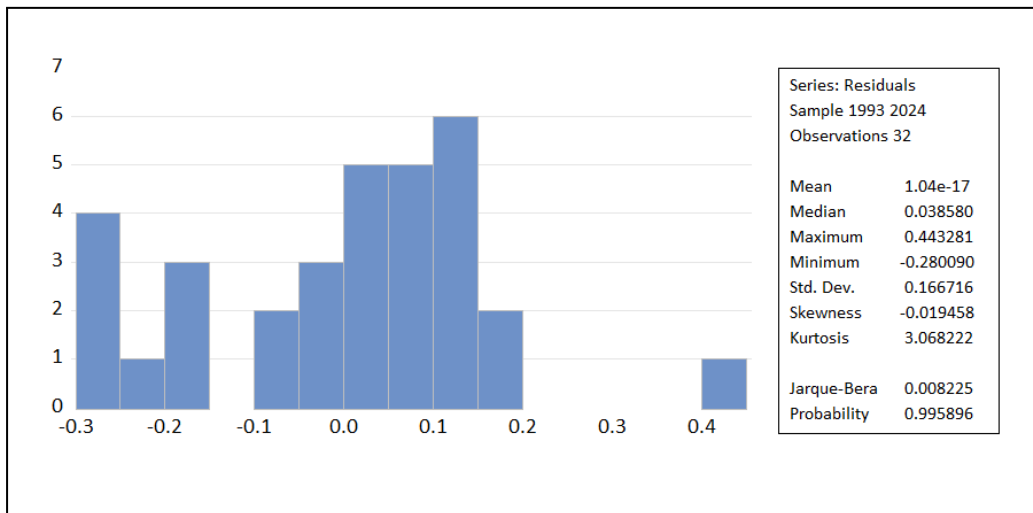
Date: 09/27/26 Time: 11:07

Sample (adjusted): 1993 2024

Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 0.114	0.114	0.4557	
		2 -0.052	-0.065	0.5522	
		3 -0.109	-0.096	0.9941	0.319
		4 0.090	0.113	1.3069	0.520
		5 -0.068	-0.107	1.4923	0.684
		6 -0.087	-0.070	1.8113	0.770
		7 -0.283	-0.262	5.3066	0.380
		8 -0.015	0.014	5.3163	0.504
		9 -0.143	-0.197	6.2809	0.507
		10 -0.118	-0.155	6.9637	0.541
		11 -0.076	-0.053	7.2625	0.610
		12 -0.109	-0.261	7.9143	0.637
		13 0.022	-0.022	7.9414	0.719
		14 0.113	-0.060	8.7097	0.728
		15 -0.032	-0.182	8.7744	0.790
		16 0.157	0.078	10.452	0.728

*Probabilities may not be valid for this equation specification.



El test de normalidad de residuos de Jarque-Bera para el AR(2) indica que NO se puede rechazar la hipótesis nula de que los residuos se comportan de manera normal.

Al estimar varios modelos encuentro que hay muchos que tienen sus coeficientes muy significativos y que tienen residuos ruido blanco, por lo que miro el test de raíz unitaria con intercepto y tendencia, para verificar si es necesario diferenciar una vez la serie.

Null Hypothesis: LN PRECIO INTERNACIONAL REAL GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 1 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-4.669099	0.0038
Test critical values:	1% level		-4.273277	
	5% level		-3.557759	
	10% level		-3.212361	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN PRECIO INTERNACIONAL REAL GH)				
Method: Least Squares				
Date: 09/26/26 Time: 21:04				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN PRECIO INTERNACIONAL REAL	-0.852434	0.182569	-4.669099	0.0001
D(LN PRECIO INTERNACIONAL REAL	0.790799	0.186325	4.244198	0.0002
C	6.161811	1.326327	4.645770	0.0001
@TREND("1991")	0.021405	0.005066	4.224895	0.0002
R-squared	0.500556	Mean dependent var		0.049230
Adjusted R-squared	0.447044	S.D. dependent var		0.235903
S.E. of regression	0.175420	Akaike info criterion		-0.526804
Sum squared resid	0.861616	Schwarz criterion		-0.343587
Log likelihood	12.42887	Hannan-Quinn criter.		-0.466073
F-statistic	9.354123	Durbin-Watson stat		1.543643
Prob(F-statistic)	0.000190			

El ADF test en niveles rechaza claramente la hipótesis nula de raíz unitaria con un p-valor de 0.0038 a un nivel de significancia del 1%.

Null Hypothesis: LN PRECIO INTERNACIONAL REAL GH is stationary				
Exogenous: Constant, Linear Trend				
Bandwidth: 6 (Newey-West automatic) using Bartlett kernel				
				LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic				0.082165
Asymptotic critical values*:				
1% level				0.216000
5% level				0.146000
10% level				0.119000
*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)				
Residual variance (no correction)				0.051848
HAC corrected variance (Bartlett kernel)				0.051097
KPSS Test Equation				
Dependent Variable: LN PRECIO INTERNACIONAL REAL GH				
Method: Least Squares				
Date: 09/28/26 Time: 14:24				
Sample: 1991 2024				
Included observations: 34				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.244111	0.078761	91.97627	0.0000
@TREND("1991")	0.024777	0.004103	6.038808	0.0000
R-squared	0.532623	Mean dependent var		7.652925
Adjusted R-squared	0.518017	S.D. dependent var		0.338076
S.E. of regression	0.234709	Akaike info criterion		-0.003915
Sum squared resid	1.762831	Schwarz criterion		0.085871
Log likelihood	2.066559	Hannan-Quinn criter.		0.026704
F-statistic	36.46720	Durbin-Watson stat		0.998569
Prob(F-statistic)	0.000001			

El KPSS test refuerza el resultado obtenido con el ADF diciendo que no hay evidencia para decir que la serie NO es estacionaria.

- Con ambos test se puede concluir que no es necesario diferenciar la serie, y que el modelo adecuado, siguiendo el criterio de información de Schwarz, sin embargo para eso necesito que tanto el MA(1) como el AR(2) tengan la misma cantidad de observaciones muestrales, y como el AR(2) tiene dos periodos de desfase debo estimar nuevamente el MA(1) con dos observaciones menos.

Dependent Variable: LN PRECIO INTERNACIONAL REAL GH				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/28/26 Time: 15:59				
Sample: 1993 2024				
Included observations: 32				
Failure to improve likelihood (non-zero gradients) after 13 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1992				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	7.221557	0.129807	55.63298	0.0000
@TREND	0.027057	0.006458	4.189433	0.0002
MA(1)	0.752803	0.136645	5.509202	0.0000
R-squared	0.674389	Mean dependent var	7.679072	
Adjusted R-squared	0.651933	S.D. dependent var	0.330916	
S.E. of regression	0.195231	Akaike info criterion	-0.340208	
Sum squared resid	1.105338	Schwarz criterion	-0.202795	
Log likelihood	8.443321	Hannan-Quinn criter.	-0.294659	
F-statistic	30.03171	Durbin-Watson stat	1.361446	
Prob(F-statistic)	0.000000			
Inverted MA Roots	-0.75			

Ahora si ambos modelos son comparables, la nueva estimación de MA(1) tiene coeficientes significativos y residuos ruido blanco.

Comparo Schwarz:

MA(1) = -0.20

AR(2) = -0.34

- Por lo que el modelo elegido es el **AR(2)** que posee un intercepto (7.24) y una tendencia determinística (2.51%). y posee raíces invertidas de 0.89, que podemos decir que son distintas de 1 por el test de Dickey-Fuller.